Integrated transportation and power system modeling

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***Abstract*—Undesired impacts on the power system caused by electric vehicles (EVs) remain a major challenge that needs to be addressed to guarantee their safe widespread adaption. In this context, efficiently balancing power demand power grids represents a fundamental issue of vehicle-to-grid (V2G) applications of EV. Here, existing approaches cover only a limited set of features required to achieve integrated multi-objective transportation and power system modeling. In this paper, we investigate an approach for holistic and integrated modeling of transportation systems and electric networks, covering electric devices within the power system and individual cars on the traffic network which make up the transportation system. The approach allows for rapid variation and evaluation of scenarios of both transportation systems and electric networks by varying their configuration, composition and underlying objectives. We apply our approach to different incremental small-scale V2G example scenarios. Obtained simulation results show the feasibility of the approach in terms of the employed modeling technique. Based on simulation results, we conclude with an evaluation of our approach with respect to it's validity, novelty, efficiency and applicability.**

***Keywords*—Feasibility study; intelligent transportation systems; smart grids; electric vehicles; rapid prototyping**

I. Introduction

In recent years increasing environmental awareness has sparked interest in electric vehicles (EVs) as one possible means to reduce dependence on fossil fuels. However, up until now undesired impacts on the power system remain a major challenge that needs to be addressed before their safe widespread adoption. In this context, inherent complexity is caused by the broad range of constraints, objectives and design alternatives both for the transportation and the power system as well as their interaction. For example, vehicle-to-grid (V2G) is a concept that considers releasing energy from EV batteries to the power system during times of increased power demand. Resulting benefits can be argued both for the transportation and the power system [1]. While major benefits include efficiency, stability and reliability of the power system, major impediments encompass faster battery degradation, need for intensive communication and infrastructure changes as well as general social, political, cultural and technical obstacles [2]. Still, possible impacts have been shown to comprise significant economic potential [3], [4]. Despite technical challenges, practice shows that today the majority of EV users (i.e. about 80%) favor charging their vehicles at home [5], which may present an acceptance problem for V2G. More importantly, the V2G example manifests the strong difference between objectives of the power system on the one side and the transportation system on the other side including their respective stakeholders. Therefore, when engineering future transportation and power systems it is important to understand their specific constraints and objectives as well as their interactions and impacts on each other. Moreover, we believe it is beneficial to estimate possible impacts as early as possible such that costly design mistakes can be avoided.

Problem: Our research focuses on methods and tools, which allow one to estimate possible impacts arising from integrating transportation and power systems as early as possible in the design process. Hereby, we are interested in microscopic effects that can be observed on the individual car and electric device level as well as the distribution network (i.e. low and medium voltage network). However, at this stage we consider approximate physical state and dynamics to be sufficient.

Contribution: We extend an existing approach to transportation system modeling and scenario exploration to include a suitable representation of the power system. For this, we propose a modeling technique allowing us to model dynamic interactions between transportation and power systems. Furthermore, we develop an approach allowing us to rapidly configure different scenarios within the transportation and the power system. Finally, we demonstrate the approach using a set of examples and discuss its current strengths and weaknesses.

Outline: First, we summarize related work and remaining deficiencies in Sec. II. Then, we describe the underlying modeling technique of our approach in Sec. III. Subsequently, we explain our approach to integrated transportation and power system modeling in Sec. IV. Thereon, for evaluation we demonstrate four basic scenarios in Sec. V. In Sec. VI we provide an extensive discussion. Finally, we conclude in Sec. VII.

II. Related work

We draw related work mainly from the fields of *modeling and simulation* as well as *optimal control* of transportation and power systems. Due to space limitations here we concentrate on approaches already integrating both domains.

In the field of *modeling and simulation* SUMO [6] has seen widespread attention for microscopic traffic simulation and its straightforward formulation and evaluation of large-scale traffic scenarios. In the context of EVs, an extension of SUMO with electrical and mechanical traction has been proposed in [7]. Kurcveil et al. [8] propose an implementation of an energy model and a charging infrastructure in SUMO. The approach allows one to model representative EV traffic scenarios in SUMO. Soares et al. [9] propose an EV scenario simulator which enables the definition of EV scenarios in the context of smart grids and distribution networks. However, its scope is limited to the definition of EV scenarios and relies on external tools for specific analysis and determining impact on the power grid. Furthermore, the height profile, i.e. the elevation slope, of routes is not considered in their energetic profile. However, increased validity of this approach has been proposed through generating realistic traffic scenarios [10].

In the field of *optimal control* one can observe a strong focus on achieving optimal charging decisions for EVs with regard to the power system. Sundstrom et al. [11] consider the capacity of the electric network during charging of EVs, thereby avoiding network overload. Schlote et al. [12] implement load-balancing strategies for charging demand at charging points into the routing decisions of EVs. Results showed reduced travel times and congestion within the employed traffic network. Deliami et al. [13] propose a real-time smart load management, which considers random plug-in of plug-in EVs (PEVs) within smart grids, considering scenarios with intermittent charging of PEVs. Alonso et al. [14] examine optimal charging scheduling for EVs in smart grids, achieving optimal behavior for charging EVs in an electric network based on a representative low-voltage network topology. Their objective is to obtain daily optimal scheduling of EV demand on transformer substations. Zakariazadeh et al. [15] propose a multi-objective scheduling approach for EVs in smart distribution networks, which considers economic and environmental objectives as well as technical constraints of distribution networks.

In summary, we found that existing approaches cover only a limited set of features required to achieve our objective (i.e. rapid multi-objective transportation and power system scenario modeling and evaluation). Both modeling/simulation and optimal control models provide sufficient representation of the transportation system while lacking detail about the power system (in particular electric devices). Additionally, the wide variety of transportation and power objectives as well as their relative importance and impacts are not addressed sufficiently.

III. Extended systems modeling technique

Our approach to rapid multi-objective transportation and power system scenario modeling and evaluation is based on emergent property estimation techniques [16] that have been integrated into a rapid prototyping method for the smart energy systems domain [17]. An instantiation of the underlying systems modeling technique for the intelligent transportation systems (ITS) domain was proposed in [18], which enables the formulation of multi-objective traffic flows as an optimal control problem over state variables (e.g. vehicle position), control variables (e.g. vehicle speed and route), constraints (e.g. collision avoidance) and objectives (e.g. energy efficiency). Finally, the approach employs a generic solver that utilizes stochastic (i.e. Monte-Carlo) optimization techniques.

Originally, the modeling technique allowed one to express static interaction between components (e.g. vehicles and controllers) based on input/output (I/O) ports (e.g. vehicle speed) and channels (e.g. vehicle speed from controller to vehicle and vice versa). In this work we extend the underlying systems modeling technique to cope with changing scenario configurations as well as the dynamic interaction between transportation and power system (i.e. vehicle and charging station) as shown in Figure 1.

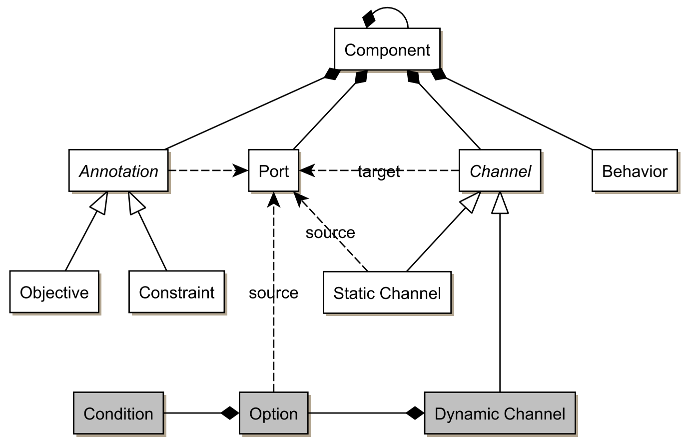


Fig. 1: Meta-model of the underlying systems modeling technique

The core element of the underlying systems modeling technique is the *component* (e.g. vehicle and charging station). Components can be either atomic (i.e. no subcomponent) or composite (i.e. one or more subcomponents) depending on their complexity. Components can be configured in their specific occurrence (e.g. vehicle length and weight). Possible interactions between components can be specified in terms of *ports* and *channels*. Ports can be either input (e.g. vehicle speed) or output (e.g. vehicle position) with respect to some component (e.g. vehicle). In contrast, channels specify the flow of information between ports. We distinguish between *static* and *dynamic channels*. Static channels define information flow from a fixed source to a fixed target port. Instead, dynamic channels define information flow from a number of possible source ports to a fixed target port. The source ports are modeled in terms of an ordered list of *options* and *conditions*. During simulation the first option is selected whose condition is satisfied. Finally, component *behaviors* and *annotations* can be specified. Behaviors calculate the component state and information flow during simulation. Annotations defined the *constraints* and *objectives* of the optimization problem instead. Thereby, annotations refer to respective boolean (for constraints) and numeric (for objectives) output ports.

IV. Transportation and power systems modeling

Based on the underlying systems modeling technique (see Sec. III), we propose an integrated transportation and power system model as shown in Figure 2. At its root, the model defines an intelligent transportation and power system component, which contains components for the transportation and the power system. The transportation system component comprises the individual car components within the traffic infrastructure. Similarly, the power system component includes individual electric components, such as static loads, solar panels, power batteries and charging stations as well as electric infrastructure consisting of low-voltage and medium-voltage net components.

Intelligent transportation and power system (IS): The IS contains the transportation system TS and the power system PS. The IS defines dynamic channels between the car load outputs of the TS and charging station load inputs of PS. Consequently, each car is connected to each charging station statically, but loads are forwarded only to connected charging stations dynamically. Therefore, the dynamic channel conditions test whether a dynamic car position matches a static charging station position. The IS specifies a cost output aggregating the cost of the TS and the PS using individual weights. On this cost output a minimization objective is defined.

Transportation system (TS): The TS comprises the traffic infrastructure TN as well as the car components C. The TN is modeled as directed graph $G = (V, E)$ with nodes V and edges $E \subseteq V \times V$. Here, nodes $v, w \in V$ represent reference points in TN such as intersections. Nodes define an absolute position in real-world coordinates with $latitude : V \to \mathbb{R}_0^+$, $longitude : V \to \mathbb{R}_0^+$ and $elevation : V \to \mathbb{R}_0^+$. In contrast, edges $(v, w) \in E$ represent road segments between source nodes v and target nodes w with a defined number of lanes as well as road type (e.g. residential streets or stations). Finally, the TS defines a cost output which aggregates the cost of the individual cars C. To ensure valid behavior, the TS implements a constraint over the car positions which calculates the largest number of pairwise overlapping cars per segment $(v, w) \in E$ and compares it to the available number of lanes.

Car (C): C represent entities on the traffic infrastructure, whose basic physical properties are length and weight. When it's departure time (a specific time point) has passed during simulation, the C starts traveling from its origin to a specified destination position. Origin, destination and position of C are edges $(v, w) \in E$ of the traffic infrastructure TN. The length of an edge is defined as $distance : V \times V \to \mathbb{R}_0^+$ between two nodes. When starting to travel, the position of a C equals the origin, while subsequent positions are selected from a set of possible routes. The set of possible routes equals the $k \in \mathbb{N}$ shortest paths from the position to the destination position or the nearest charging station. Of these alternatives one route is selected randomly. The current position consists of the current edge e and traveled distance d on this edge as $position = \{(e, d) : E \times \mathbb{R}_0^+\}$ with $e = (v, w)$ such that $d \leq distance(v, w)$. C contain a battery with a state of charge (SOC) and a defined capacity to which the C's energy consumption and production are applied to. Energy consumption occurs during driving along the TN or during discharge at charging stations. While driving, energy consumption is based on the C's energy efficiency, traveled elevation profile, weight as well as the randomly selected speed (limited to a maximum) in each simulation step. The traveled elevation profile is defined as $slope : V \times V \to \mathbb{R}_0$. Energy production occurs during driving (i.e. energy recuperation) or during charging at charging stations. To ensure valid behavior, a constraint

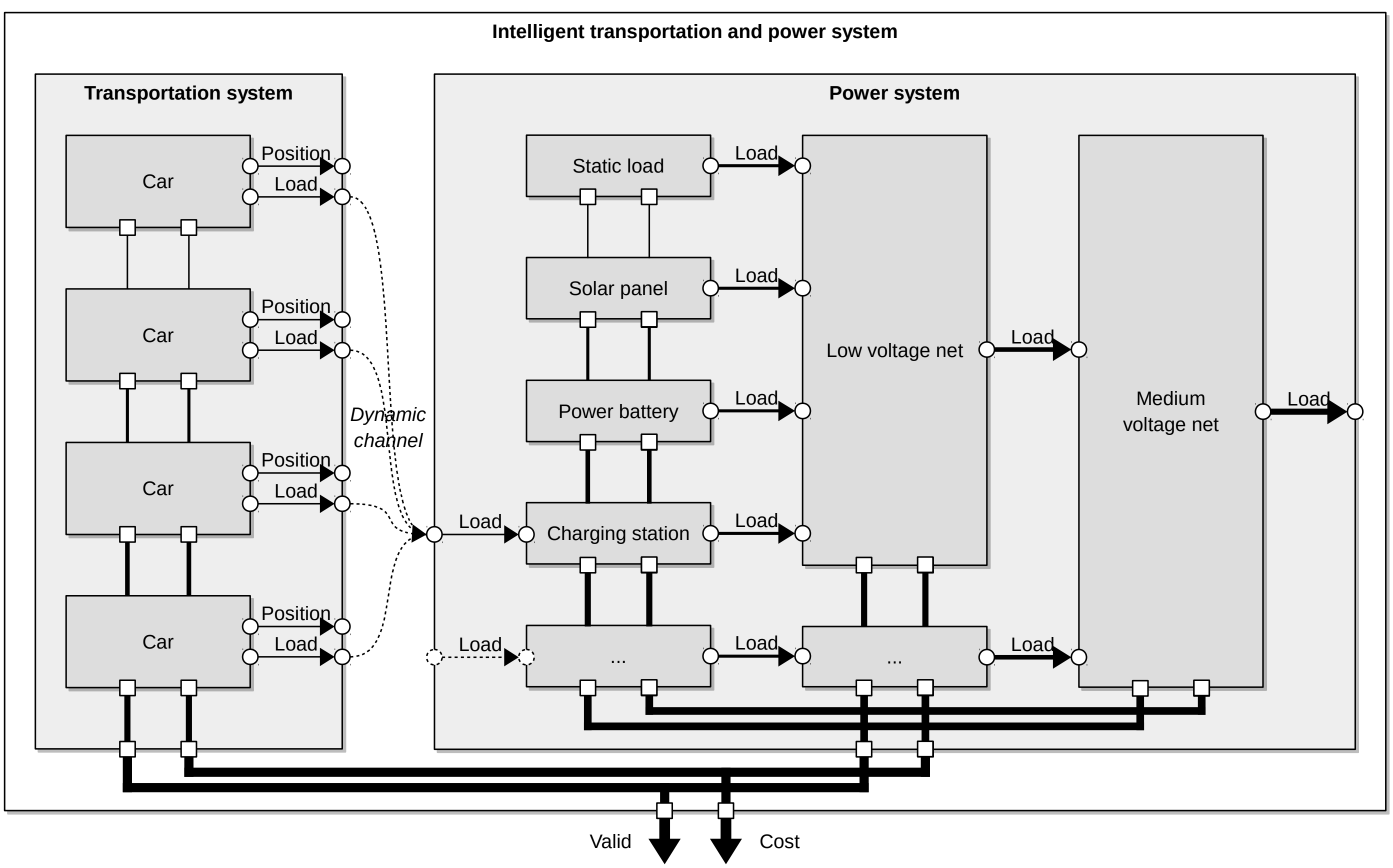


Fig. 2: Overview of the holistic modeling approach including transportation and power system connected by dynamic channels.

tests whether the SOC lies within a constant minimum and maximum. When connected to a charging station, a charging mode is chosen randomly. If the C chooses charging, a load is transferred from the charging station to the C. If the C chooses discharging, a load is transferred from the C to the charging station instead. Furthermore, the C can choose an idle state. Finally, the C objective includes multiple cost factors. The first cost factor evaluates the C's state of charge with respect to the C's maximum state of charge. The second cost factor aggregates the time needed to reach its specified destination position. The third cost factor measures the energy consumption in relation to the maximum energy consumption. Then, the cost factors are aggregated and weighted using the weights of the individual cost factors.

Power system (PS): The PS represents the overall electric infrastructure including individual electric devices as well as low-voltage nets LV and medium-voltage nets MV (higher voltage nets are omitted currently). Currently supported devices include static loads SL, solar panels SP, power batteries PB and charging stations CS. At its interface the PS specifies a cost output that aggregates the cost of the electric infrastructure and it's devices. The PS forwards the load of the medium-voltage net representing the final load balance.

Low/medium-voltage nets (LV/MV): LV/MV receive loads from connected electric devices and provide an aggregate load output to the upper voltage level. Note that connected electric devices can be LV as well. Based on the aggregated load, a net's cost output is specified relative to the total net capacity (i.e. the closer to the capacity, the worse). Then, each net's cost output is dampened based on a defined weight. Finally, to ensure valid behavior a constraint tests if a net's aggregated load as well as all connected loads lie within the net's total capacity. More information on the model can be found in [19].

Static load (SL): SL aggregate a wide range of electric devices, which cause non-controllable loads (such as conventional light bulbs or washing machines). Consequently, they abstract from specific details about different electric devices, i.e. producers and consumers, reducing the available information to their load per time instant only. The result of this aggregation is defined in terms of a load profile $L = (l_t)_{0 \leq t \leq n}$ with $t, n \in \mathbb{N}$ representing current and maximum time points and $l_t \in \mathbb{R}$ representing positive or negative loads.

Power battery (PB): PB represent stationary electric devices acting as both producers and consumers (e.g. lead-acid batteries). Depending on random selection of a mode in each simulation step, energy consumption or production can be observed on the low-voltage net side. Production and consumption are subject to a defined charge rate and affect the state of charge (SOC) of the PB which has a specified capacity. To ensure valid behavior, a constraint tests whether the SOC lies within a constant minimum and maximum. Detailed information about the model can be found in [17].

Solar panel (SP): SP are electric devices which represent one class of renewable energy producers with a defined yield. Based on a specific power scale, SP produce a positive load peaking at the mean time point with a specified variance. Based on random selection SP can choose to dampen their production (known as maximum power point tracking). Detailed information about the model can be found in [17].

Charging station (CS): CS represent electric devices acting as consumers or producers and interacting with the cars C of the transportation system TS. They are assigned a physical location, i.e. an edge $(v, v) \in E$ of the traffic network TN with zero length (i.e. same source and target node $v \in V$), representing the $position : CS \rightarrow E$. Based on dynamic connection to the cars, CS behavior is determined by the selected charging mode of the currently connected car C. When a load is transferred from the CS to the car, energy consumption can be observed on the low-voltage net side. In contrast, when a load is transferred from the car to the CS, energy production can be observed. Transfer rates are subject to a specific charge rate.

V. DEMONSTRATION

To demonstrate the presented approach, we evaluate four example scenario configurations. Generally, all examples are simulated with a time resolution of 60 seconds per simulation step and 30 simulation steps in total, aggregating to a total duration of 30 minutes. The examples evaluate the effects of varying the system objective focus (weight) between the transportation system cost and the power system cost as well as the state of expansion of the power system.

Within the power system PS, all scenarios include ten static load components SL, five solar panel components SP and five power battery components PB with a specific reference configuration and stage of expansion. In all scenarios charging stations as well as four low-voltage nets LV and one medium-voltage net MV with constant reference configurations are employed. The stage of expansion of the power system is varied from Scenarios 1-2 to Scenarios 3-4, where solar panel capacities are increased twofold and battery capacities are increased fourfold. Furthermore, in Scenarios 1-2 16 charging stations are employed, in contrast to 56 charging stations in Scenarios 3-4.

In terms of the transportation system TS, the examples feature a total number of 440 cars C and a traffic network TN of 45 kilometers in length and 45 kilometers in width. The cars are distributed equally among three configurations, which differ in the cars' initial state of charge (SOC) with 33%, 66% and 100% of the maximum SOC respectively. The origins of cars are drawn randomly from all edges E of the TN. Furthermore, the destinations are drawn randomly from a set of the most outward edges of TN. Finally, for origins we prefer the lower left sector of TN, while for destinations we prefer the upper right.

The results of control optimization are shown in Tab. I. For each scenario, a traffic flow graph and power chart is provided as well as general statistics and performance characteristics.

Scenario 1: Prefer TS, low stage of expansion: In terms of traffic flows, results show high frequency of routes utilized representing shortest paths to destinations. Net balance is prone to high fluctuations in load as little equalization of net balances is performed by cars charging/discharging at charging stations.

Scenario 2: Prefer PS, low stage of expansion: Results show traffic flows with a higher frequency of edges visited around and on charging stations. Furthermore, compared to Scenario 1, cars utilize shortest path routes to destinations less frequently, which results in marginally higher total distance traveled by cars. Also, net balance is more equalized with cars intermittently discharging at a low number of charging stations.

Scenario 3: Prefer TS, high stage of expansion: Traffic flows show a high frequency of routes utilized representing shortest paths to destinations. Overall distance traveled by cars is decreased, while final distance to destination is increased. Compared to Scenario 2, a higher level of equalization of net balance can be observed, due to a higher number of charging stations enabling cars to charge/discharge and higher capacities of solar panels and power batteries.

Scenario 4: Prefer PS, high stage of expansion: Results show cars utilizing shortest path routes less frequently compared to Scenarios 1 and 3, resulting in higher distance traveled and final distance to destination. Compared to Scenario 2 and 3, frequency is distributed more evenly across the traffic network, resulting in higher frequency of edges visited around and on charging stations. Consequently, due to cars charging/discharging at a high number of charging stations, in contrast to Scenario 3, net balances are more equalized.

VI. Discussion

In the following we discuss our approach with respect to four questions: How *valid* are the underlying system models and behavior estimation results? How *novel* is the approach compared to existing studies on electro-mobility? How *efficient* is the technique to evaluate different scenarios? And finally how *applicable* is our approach in practice?

Validity: Our employed system modeling technique enables an early estimation of system behavior and utilizes approximate physical state and dynamics. Model state is evaluated in discrete time steps and employs discrete values for state. Model accuracy could be improved by representing state and time continuously as well as utilizing detailed physical models for transportation systems and power systems, i.e. vehicles and electric devices. However, computational complexity of such models during behavior estimation could prove to be an issue as we currently employ discrete steps to limit possible behavior space and computational complexity to acceptable levels. In the example scenarios, a model of the traffic network is considered, which neglects low-level traffic infrastructure such as traffic lights and turns within roads. To improve it's representativity, traffic networks with a level of detail of comparable real world traffic infrastructures could be employed. Furthermore, while the behavior of individual cars can be observed microscopically in our model, our vehicle model doesn't consider physically accurate measures of acceleration, deceleration, gravitational forces during turns and energy consumption. Instead, it uses approximate values for traveling speed and energy consumption during simulation steps. Additionally, some objectives of interest to the concept

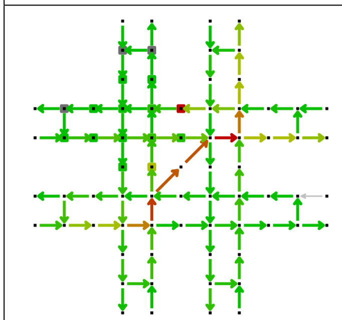
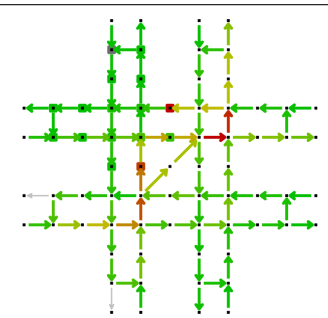
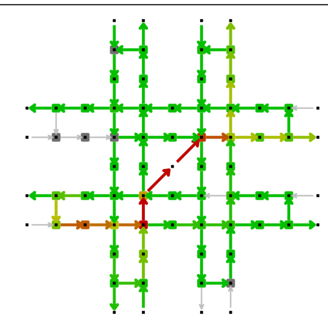
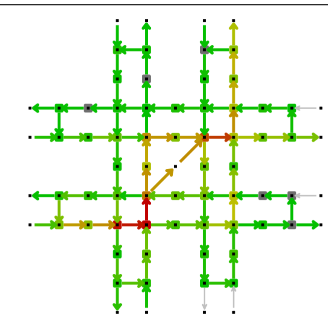
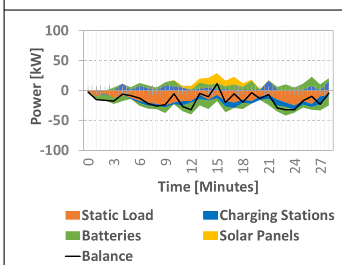
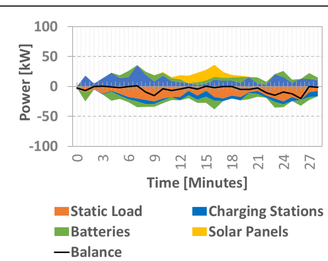
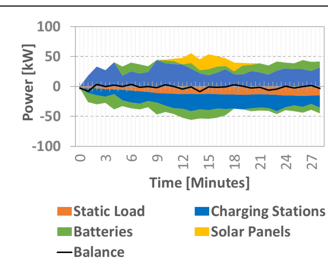
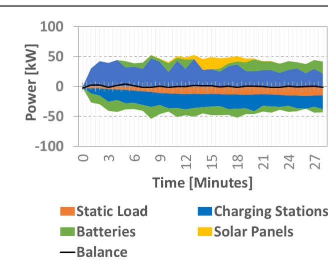
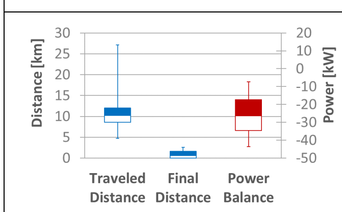
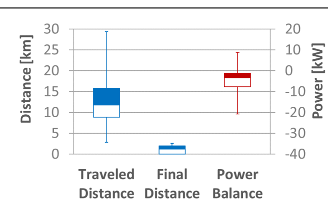
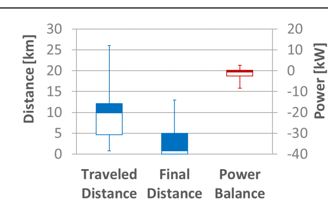
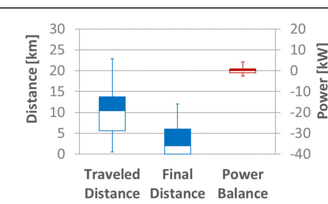
Scenario 1		Scenario 2		Scenario 3		Scenario 4	
							
							
							
Computation time	229s	*Computation time*	250s	*Computation time*	218s	*Computation time*	227s
Memory usage	6.03GB	*Memory usage*	5.27GB	*Memory usage*	6.41GB	*Memory usage*	6.75GB

TABLE I: Traffic flow graph, power chart, statistics, computation time and memory usage for each scenario.

of V2G such as minimization of EV battery degradation and optimization of EV charging/discharging decisions based on energy price have been omitted in our model. It also remains an issue whether configurations employed in our example scenarios are representative for real world scenarios. To obtain a measure of validity, the quality of results obtained with our approach has to be compared with detailed simulations of traffic and electrical networks.

Novelty: We employ a parameter-based approach allowing rapid modeling and evaluation of multi-objective transportation and power system scenarios. Compared to existing approaches outlined in Sec. II, our approach allows quick and adaptable model formulation with regard to transportation and power systems. Furthermore, our approach supports multiple objectives, which can be rapidly established or adjusted for systems or subsystems. As demonstrated in Sec. V, this enables one to rapidly evaluate scenarios concerning evolving systems, e.g. different expansion levels of renewable and smart energy devices or changing system objective focus.

Efficiency: Our approach represents a rapid prototyping technique enabling to model, estimate and evaluate transportation and power system scenarios. For this, we employ incremental techniques allowing model refinement through multiple iterations. Here, the efficiency of our approach can be measured in terms of lines of code necessary to establish a specific scenario. Reducing time and resource intensive modeling, in the presented example scenarios our approach enabled to quickly establish different scenarios constantly within 350 lines of code without requiring extensive adjustments.

Applicability: The practical applicability of our approach is strongly dependent on employed domain knowledge specific for transportation and power systems. Here, a key issue is the fragmentation of knowledge into different domains and the autonomous goals involved for stakeholders. Therefore, the practical applicability of our modeling technique depends on different domains collaborating and sharing their knowledge.

VII. Conclusion

In this paper, we proposed an approach to rapidly model and evaluate transportation and power system scenarios. The proposed holistic and component-based approach is able to model power systems with detailed electric devices as well as transportation systems with individual cars microscopically. Our approach allows to rapidly explore and vary configurations of transportation system and electric network scenarios. We evaluated several incremental scenarios varying in stage of expansion of renewable and smart energy devices as well as focus of system objectives. Scenario results demonstrated the feasibility of the modeling technique and rapid evaluation of multiple scenarios. However, key issues which have to be addressed are the validity and applicability the approach. Future work includes refinement of model accuracy through vehicle model improvements and utilization as well as modeling of additional objectives and systems. Furthermore, modeling representative traffic infrastructures with OpenStreetMap and validation of results obtained with our approach represent major next steps. Finally, to be able to handle larger problem scales, more detailed models and resulting computational complexity, we currently work on effective behavior estimation algorithms.

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